Pascal's Formula I

- The binomial coefficients $\binom{n}{k}$ has been defined for all non-negative integers k and n. If n is positive and $1 \leq k \leq n$,

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n(n-1)\cdots(n-k+1)}{k(k-1)\cdots 1}. \tag{5.1}$$

We have the relation

$$\binom{n}{k} = \binom{n}{n-k}$$

for integers all k and n with $0 \leq k \leq n$.

Theorem

Pascal's formula – For all integers n and k with $1 \leq k \leq n-1$,

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}.$$

Pascal's Formula II

Pascal's triangle.

$n\backslash k$	0	1	2	3	4	5	6	7	8	$\cdots$
0	1									
1	1	1								
2	1	2	1							
3	1	3	3	1						
4	1	4	6	4	1					
5	1	5	10	10	5	1				
6	1	6	15	20	15	6	1			
7	1	7	21	35	35	21	7	1		
8	1	8	28	56	70	56	28	8	1	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

- Many of the relations involving binomial coefficients can be discovered by carefully examination of Pascal's triangle.
- The symmetry relation

$$\binom{n}{k} = \binom{n}{n-k}$$

is readily noticed.



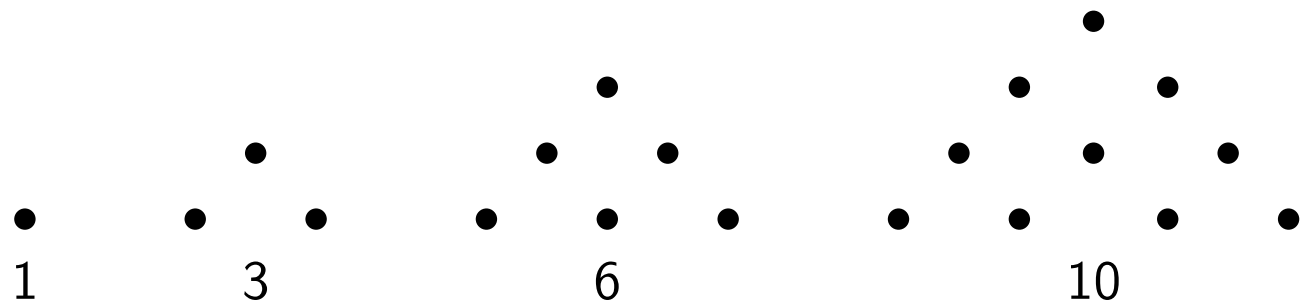
Pascal's Formula III

- The identity

$$\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n$$

is discovered by adding the numbers in a row of Pascal's triangle.

- The numbers $\binom{n}{1} = n$ in column $k = 1$ are the counting numbers. The numbers $\binom{n}{2} = n(n-1)/2$ in column $k = 2$ are the so-called *triangular numbers*, which equal to the number of dots illustrated in the following figure.



- The numbers $\binom{n}{3} = \frac{n(n-1)(n-2)}{3 \cdot 2 \cdot 1}$ in column $k = 3$ are the so-called *tetrahedral numbers*.



Pascal's Formula IV

- Define $p(n,k)$ as the number of paths from the top left corner (the entry $\binom{0}{0}=1$) to the entry $\binom{n}{k}$ where in each path we move from one entry to the entry in the next row immediately below it or immediately to its right.
- We define $p(0,0)$ to be 1. and, for each nonnegative integer n, we have

$$p(n,0)=1, \qquad \text{we must move straight down to reach } \binom{n}{0}$$

and

$$p(n,n)=1 \qquad \text{we must move diagonally to reach } \binom{n}{n}.$$

Each path from $\binom{0}{0}$ to $\binom{n}{k}$ is either

1. a path from $\binom{0}{0}$ to $\binom{n-1}{k}$ followed by one vertical move or
2. a path from $\binom{0}{0}$ to $\binom{n-1}{k-1}$ followed by one diagonal move.



Pascal's Formula V

Thus, we have

$$p(n,k) = p(n-1,k) + p(n-1,k-1),$$

a Pascal-type relation for the numbers $p(n,k)$. Hence it follow that for all integers n and k with $0 \leq k \leq n$,

$$p(n,k) = \binom{n}{k}.$$



The Binomial Theorem I

Theorem

Let n be a positive integer. Then, for all x and y,

$$(x+y)^n = x^n + \binom{n}{1}x^{n-1}y + \binom{n}{2}x^{n-2}y^2 + \cdots + \binom{n}{n-1}xy^{n-1} + y^n.$$

In summation notation,

$$(x+y)^n = \sum_{k=0}^{n}\binom{n}{k}x^{n-k}y^k.$$

The binomial theorem can be written in several other equivalent forms

$$(x+y)^n = \sum_{k=0}^{n}\binom{n}{n-k}x^{n-k}y^k,$$



The Binomial Theorem II

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{n-k} x^k y^{n-k},$$

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.$$

Theorem

Let n be a positive integer. Then, for all x,

$$(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k = \sum_{k=0}^{n} \binom{n}{n-k} x^k.$$



Identities I

$$k\binom{n}{k} = n\binom{n-1}{k-1}, \quad (n \text{ and } k \text{ are positive integers}) \tag{5.2}$$

follows immediately from the fact that $\binom{n}{k} = 0$ if $k > n$ and

$$\binom{n}{k} = \frac{n(n-1)\cdots(n-k+1)}{k(k-1)\cdots 1}$$

for $1 \le k \le n$. The other identities

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n} = 2^n, \quad (n \ge 0). \tag{5.3}$$

$$\binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \cdots + (-1)^n\binom{n}{n} = 0, \quad (n \ge 1). \tag{5.4}$$

$$\binom{n}{0} + \binom{n}{2} + \cdots = \binom{n}{1} + \binom{n}{3} + \cdots, \quad (n \ge 1). \tag{5.5}$$



Identities II

$$\binom{n}{0} + \binom{n}{2} + \cdots = 2^{n-1}, \tag{5.6}$$

$$\binom{n}{1} + \binom{n}{3} + \cdots = 2^{n-1}. \tag{5.7}$$

Using identities (5.2) and (5.3),

$$1\binom{n}{1} + 2\binom{n}{2} + \cdots + n\binom{n}{n} = n2^{n-1}, \quad (n \geq 1). \tag{5.8}$$

$$n(n+1)2^{n-2} = \sum_{k=1}^{n} k^2 \binom{n}{k}, \quad (n \geq 1). \tag{5.9}$$

An identity for the sum of the squares of the numbers in a row of Pascal's triangle is

$$\sum_{k=0}^{n} \binom{n}{k}^2 = \binom{2n}{n}, \quad (n \geq 0). \tag{5.10}$$



Identities III

We now extend the domain of definition of the numbers $\binom{n}{k}$ to allow n to be any real number and k to be any integer (positive, negative, or zero). We define

$$\binom{r}{k} = \begin{cases} \frac{r(r-1)\cdots(r-k+1)}{k!} & \text{if } k \geq 1, \\ 1 & \text{if } k = 0, \\ 0 & \text{if } k \leq -1. \end{cases}$$

For instance

$$\binom{5/2}{4} = \frac{(5/2)(3/2)(1/2)(-1/2)}{4!} = \frac{-5}{128},$$

$$\binom{-8}{2} = \frac{(-8)(-9)}{2} = 36,$$

$$\binom{3.2}{0} = 1,$$

$$\binom{3}{-2} = 0.$$



Identities IV

Pascal's formula and formula (5.2), namely

$$\binom{r}{k} = \binom{r-1}{k} + \binom{r-1}{k-1}$$

and

$$k\binom{r}{k} = r\binom{r-1}{k-1}$$

are now valid for all r and k.

$$\begin{aligned}\binom{r}{0} + \binom{r+1}{1} + \binom{r+2}{2} + \cdots + \binom{r+k}{k} \\ = \binom{r+k+1}{k}.\end{aligned} \tag{5.11}$$



Identities V

Identity (5.11) is valid for all real number r and all integers k. Moreover we have

$$\begin{aligned}\binom{0}{k}+\binom{1}{k}+\cdots+\binom{n-1}{k}+\binom{n}{k}\\ =\binom{n+1}{k+1}.\end{aligned}\tag{5.12}$$

Identity (5.12) is valid for all non-negative integers k and n.



Unimodality of Binomial Coefficients I

Theorem

Let n be a positive integer. The sequence of binomial coefficients,

$$\binom{n}{0}, \binom{n}{1}, \binom{n}{2}, \ldots, \binom{n}{n}$$

is a unimodal sequence. More precisely, if n is even,

$$\binom{n}{0} < \binom{n}{1} < \cdots < \binom{n}{n/2}, \quad \binom{n}{n/2} > \cdots > \binom{n}{n-1} > \binom{n}{n},$$

and if n is odd,

$$\binom{n}{0} < \binom{n}{1} < \cdots < \binom{n}{(n-1)/2} = \binom{n}{(n+1)/2} > \cdots > \binom{n}{n-1} > \binom{n}{n}.$$



Unimodality of Binomial Coefficients II

Corollary

For n a positive integer, the largest of the binomial coefficients

$$\binom{n}{0}, \binom{n}{1}, \binom{n}{2}, \ldots, \binom{n}{n}$$

is

$$\binom{n}{\lfloor n/2 \rfloor} = \binom{n}{\lceil n/2 \rceil}.$$

Definition

Let S be a set of n elements. A *clutter* (called an *antichain* in the more general context of a partially ordered set) of S is a collection $\mathcal{C}$ of combinations of S with the property that no combination in $\mathcal{C}$ is contained in another.



Unimodality of Binomial Coefficients III

Example

If $S = \{a, b, c, d\}$, then

$$\mathcal{C} = \{\{a, b\}, \{b, c, d\}, \{a, d\}, \{a, c\}\}$$

is a clutter.

One way to obtain a clutter on a set S is to choose an integer $k \leq n$ and then take $\mathcal{C}_k$ to be the collection of all k-combinations of S. Since each combination in $\mathcal{C}_k$ has k elements, no combination in $\mathcal{C}_k$ can contain another and hence $\mathcal{C}_k$ is a clutter.

Theorem

Let S be a set of n elements. Then a clutter on S contains at most $\binom{n}{\lfloor n/2 \rfloor}$ sets.



The Multinomial Theorem I

In the general formula, the role of the binomial coefficients is taken over by numbers called the *multinomial coefficients*, which are defined by

$$\begin{pmatrix} n \\ n_1 n_2 \cdots n_t \end{pmatrix} = \frac{n!}{n_1! n_2! \cdots n_t!} \tag{5.13}$$

with

$$n_1 + n_2 + \cdots n_t = n.$$



The Multinomial Theorem II

Theorem

Let n be a positive integer. For all $x_1, x_2, \ldots, x_t$,

$$(x_1 + x_2 + \cdots + x_t)^n = \sum \binom{n}{n_1 n_2 \cdots n_t} x_1^{n_1} x_2^{n_2} \cdots x_t^{n_t},$$

where the summation extends over all nonnegative integral solutions $n_1, n_2, \ldots, n_t$ of $n_1 + n_2 + \cdots + n_t = n$.

Example

When $(x_1 + x_2 + x_3 + x_4 + x_5)^7$ is expanded, the coefficient of $x_1^2 x_3 x_4^3 x_5$ equals

$$\binom{7}{2\,0\,1\,3\,1} = \frac{7!}{2!0!1!3!1!} = 420.$$



The Multinomial Theorem III

Example

When $(2x_1 - 3x_2 + 5x_3)^6$ is expanded, the coefficient of $x_1^3 x_2 x_3^2$ equals

$$\binom{6}{3\,1\,2} 2^3(-3)5^2 = -36,000.$$



Newton's Binomial Theorem I

Theorem

Let α be a real number. Then, for all x and y with $0 \leq |x| \leq |y|$,

$$(x+y)^{\alpha} = \sum_{k=0}^{\infty} \binom{\alpha}{k} x^k y^{\alpha-k},$$

where

$$\binom{\alpha}{k} = \frac{\alpha(\alpha-1)\cdots(\alpha-k+1)}{k!}.$$



Newton's Binomial Theorem II

- $\alpha = -n$

$$\begin{aligned}\binom{-n}{k} =& \frac{-n(-n-1)\cdots(-n-k+1)}{k!} \\ =& (-1)^k \frac{n(n+1)\cdots(n+k-1)}{k!} \\ =& (-1)^k \binom{n+k-1}{k}\end{aligned}$$

and

$$(1-z)^{-n} = \frac{1}{(1-z)^n} = \sum_{k=0}^{\infty} \binom{n+k-1}{k} z^k. \tag{5.14}$$



Newton's Binomial Theorem III

- $\alpha = \frac{1}{2}$

$$\begin{aligned}\binom{1/2}{k} &= \frac{\frac{1}{2}\left(\frac{1}{2}-1\right)\cdots\left(\frac{1}{2}-k+1\right)}{k!} \\ &= \frac{(-1)^{k-1}}{2^k}\frac{1\cdot 2\cdot 3\cdot 4\cdots(2k-3)(2k-2)}{k!\cdot 2\cdot 4\cdots(2k-2)} \\ &= \frac{(-1)^{k-1}}{k\cdot 2^{2k-1}}\frac{(2k-2)!}{(k-1)!(k-1)!} \\ &= \frac{(-1)^{k-1}}{k\cdot 2^{2k-1}}\binom{2k-2}{k-1}.\end{aligned}$$

and

$$(1+z)^{1/2} = 1 + \sum_{k=1}^{\infty}\frac{(-1)^{k-1}}{k\cdot 2^{2k-1}}\binom{2k-2}{k-1}z^k.$$



More on Partially Ordered Sets I

- Let $(X, \leq)$ be a finite partially ordered set. An *antichain* is a subset A of X no pair of whose elements are comparable. In contrast, a *chain* is a subset C of X each pair of whose elements is comparable.

Example

Let $X = \{1, 2, \cdots, 10\}$, and consider the partially ordered set $\{X, |\}$ whose partial order $|$ is "is divisible by". Then $\{4, 6, 7, 9, 10\}$ is an antichain of size 5, while $\{1, 2, 4, 8\}$ is a chain of size 4. There are no antichains of size 6 and no chains of size 5.

Theorem

Let $(X, \leq)$ be a finite partially ordered set, and let r be the largest size of a chain. Then X can be partitioned into r but no fewer antichains.



More on Partially Ordered Sets II

Theorem

Let $(X, \leq)$ be a finite partially ordered set, and let m be the largest size of an antichain. Then X can be partitioned into m but no fewer chains.

